

THz Biomedical Sensing for Early Cancer Detection: Metamaterial Graphene Biosensors With Rotated Split-Ring Resonators

Marwa Rezeg¹, Aymen Hlali¹, and Hassen Zairi¹, *Senior Member, IEEE*

Abstract—A highly sensitive graphene-based metamaterial biosensor is proposed and analyzed for early cancer detection. The sensor design employs three circular graphene split ring resonators to achieve polarization-insensitive performance. The finite element method simulation results confirm that the designed biosensor exhibits a tunable sensing capability. The material under test covers the surface of the biosensor, where resonance occurs with high absorption. The resonance frequencies of the sensor are dependent on the optical properties of the analyte sample, enabling the device to differentiate between various cancer cell types, including skin, blood, cervical, adrenal gland, and breast cancer. Graphene's tunability is leveraged to study the effects of chemical potential, relaxation time, and temperature, with the aim of maximizing the sensor's sensitivity. The designed biosensor, which detects variations in refractive index, exhibits a maximum sensitivity of 3.880 THz/RIU, a Q-factor of 8.948, and a figure-of-merit of 8.146 RIU⁻¹ for healthy and cancerous cell samples. The spatial patterns of electric and magnetic fields, surface current distribution, and power flow of the proposed biosensor have been thoroughly analyzed to ensure its sensitivity and suitability for biomedical applications. The results demonstrate the potential of the THz sensor for early cancer detection.

Index Terms—Graphene, metamaterial biosensor, cancer cells, split ring resonators, terahertz band.

I. INTRODUCTION

CANCER, a multifaceted array of diseases characterized by uncontrollable cell growth and proliferation, poses a significant global health challenge. As stated by the World Health Organization (WHO), cancer stands as the second highest contributor to global human mortality, responsible for an estimated 20 million cases and 10 million deaths in 2022 alone [1]. Delayed diagnosis and treatment are significant factors contributing to the high mortality rate of cancer. Cancer often remains asymptomatic in its early stages, making it challenging to detect and treat before it progresses to more advanced stages. This delay in treatment can lead to poorer prognosis and reduced survival rates, emphasizing the importance of timely intervention. To address the issue of treatment delays, a multifaceted approach

is necessary. This approach should encompass early detection initiatives, improved access to healthcare, technological innovation, and robust support systems for patients.

The progress in nanoscience and material innovation has paved the way for the creation of point-of-care devices designed for in situ and portable sensing, with the potential to improve patient outcomes and lower mortality rates [2]. The emergence of label-free biosensors, which are highly sensitive biomolecule detection sensors, has led to the development of ideal tools for cancer diagnosis [3], [4]. High sensitivity and specificity are essential criteria for these biosensors to effectively detect cancer biomarkers, providing accurate and reliable results in a compact and portable device [5], [54], [55]. Also, the biosensor should be designed to operate in real-time with minimal harm to body tissue, thereby reducing the risk of injury and adverse effects [6].

Terahertz radiation, spanning frequencies between microwave and infrared wavelengths and ranging from 100 GHz and 10 THz, offers unique capabilities for imaging, spectroscopy, and communication applications [7], [56], [57]. However, it is still technologically underdeveloped. As researchers become more familiar with terahertz radiation, there is a rising demand for advanced terahertz equipment that can provide enhanced sensing capabilities. This push for innovation in terahertz technology is driven by the need for high-performance sensing tools that can address the complexities of modern healthcare challenges. Terahertz technology offers a wide array of applications across diverse fields, including security [8] and materials science [9], as well as communications [10], medicine [11] and energy harvesting [61]. Considering all these examples, terahertz (THz) absorbers are a highly attractive and practical application within the THz range. Recent advancements in this field have led to the design of absorbers using metamaterial layers, which provide the capability to control the properties of electromagnetic waves in unprecedented ways.

The remarkable characteristics of metamaterials are a result of their internal physical structure, distinguishing them from natural materials whose properties are primarily defined by their chemical composition and bonding. The exceptional impact of metamaterials on the transmission of light through them is the key factor driving the intense interest in this field. These materials consist of subwavelength structures arranged in periodic or random patterns, which give rise to unique electromagnetic (EM) properties that are not found in natural materials [12],

Manuscript received 13 May 2024; revised 5 June 2024; accepted 10 June 2024. Date of publication 12 June 2024; date of current version 20 June 2024. (Corresponding author: Marwa Rezeg.)

The authors are with the Research Laboratory Smart Electricity & ICT, SEICT, National Engineering School of Carthage, University of Carthage, Tunis 1054, Tunisia (e-mail: marwa.rezeg@enicar.ucar.tn).

Digital Object Identifier 10.1109/JPHOT.2024.3413528

[58] such as negative refractive index (RI), invisibility cloaking, negative permeability, and artificial magnetism.

Negative refractive index (RI) is an intriguing property that is not found in natural materials. This property arises when both the imaginary part of electric permittivity (ϵ) and magnetic permeability (μ) are negative. This phenomenon has been demonstrated in various metamaterial structures, including splitting resonators made with graphene [13], [14]. By periodically arranging graphene layers, researchers can create metamaterial structures that exhibit a negative refractive index, among other unique electromagnetic properties. The distinctive characteristics of graphene, such as its high carrier mobility, tunable conductivity [62], and strong light-matter interaction, make it an ideal candidate for metamaterial applications [15], [16], [17].

The near-perfect absorption properties of metamaterial absorbers have garnered considerable attention due to their exceptional physical properties. The ultra-thin dielectric layer depth of these absorbers allows for a significant reduction in the overall thickness of the device [18]. The adaptability of metamaterial absorbers to non-uniform surfaces is a significant advantage, as it enables their integration into complex and irregularly shaped structures [19]. Additionally, the versatile structure design of these absorbers allows for the tailoring of their absorption properties to specific frequency ranges [20], [59].

One area where metamaterial absorbers hold significant promise in sensing is in the development of highly sensitive terahertz (THz) sensors. The resonant peak frequency of these absorbers responds to variations in the refractive index (RI) of the tested medium (analyte) with elevated accuracy and sensitivity [21], [22]. By incorporating graphene into metamaterial absorbers, researchers can exploit its properties to enhance the absorption of THz radiation and improve the sensor's detection capabilities. Graphene consists of a single layer of carbon atoms arranged in a hexagonal lattice, forming a two-dimensional material [23]. It has high carrier mobility, tunable conductivity [60], and strong light-matter interaction [24].

In the field of health care, THz radiation has been explored for the detection and monitoring of various types of cancer, including skin cancer [25], breast cancer [26], [27], colon cancer [28], blood cancer [29], cervical cancer [30], and adrenal cancer [31]. The high absorption coefficient of biological cells in the THz range facilitates the detection of subtle changes in cellular properties, enabling the early identification of cancerous growths. THz radiation has also been used for the diagnosis and monitoring of non-cancerous health conditions, such as psoriasis, eczema, dermatitis in dermatologic health, bone health, dental health, detection of atherosclerosis and other cardiovascular conditions, and drug delivery monitoring.

The non-ionizing nature of THz radiation makes it a safer and more comfortable option for patients compared to mammography, which uses ionizing radiation [32]. Additionally, THz radiation can penetrate several millimeters of tissue, allowing for the imaging of internal structures without the need for invasive procedures [33].

This paper proposes an innovative THz Metamaterial sensor that is designed and analyzed for detecting cancer cells. The sensitivity of the THz Metamaterial sensor is predicted by analyzing

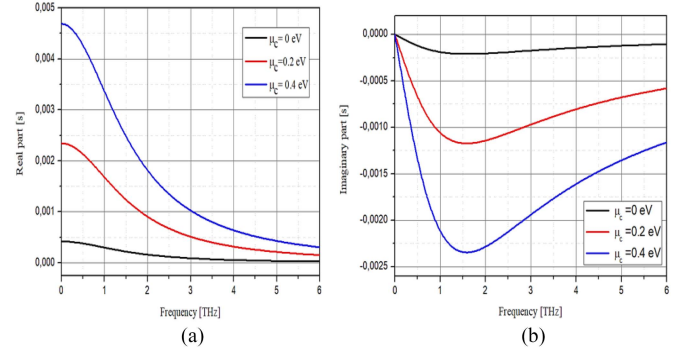


Fig. 1. The conductivity of graphene for various chemical potentials.

the refractive index profile of various cancer cells, and the results are reported. To improve healthcare outcomes, the integration of terahertz Metamaterials, graphene, and microscale technology is explored, aiming to increase the sensor's sensitivity, selectivity, and overall diagnostic accuracy.

II. THEORY AND FORMULATION

A. Graphene Conductivity

Graphene's surface conductivity is a crucial parameter that determines its electrical properties and potential applications. The surface conductivity of graphene is described in terms of $\sigma_g(\omega, \mu_c, \Gamma, T)$, where ω is the angular frequency, μ_c is the chemical potential, Γ is the relaxation time, and T is the temperature [35].

The Kubo formula is a widely used theoretical framework for calculating the conductivity of graphene. The Kubo formula is based on the quantum mechanical description of the electron transport in graphene and provides a thorough comprehension of the conductivity behavior of graphene [34]. The specifications of devices based on graphene, including absorbers and sensors, can be altered by adjusting the variable parameters within σ_g . The conductivity of graphene comprises both inter-band and intra-band terms. (1) describes the surface conductivity of graphene.

$$\sigma_g(\omega, \mu, \Gamma, T) = \sigma_{\text{intra}} + \sigma_{\text{inter}} \quad (1)$$

where

$$\sigma_{\text{intra}} = -j \frac{e^2 k_B T}{\pi \hbar^2 (\omega - j2\Gamma)} \left[\frac{\mu_c}{K_B T} + 2 \ln \left(e^{\left(-\frac{\mu_c}{K_B T} \right)} + 1 \right) \right] \quad (2)$$

$$\sigma_{\text{inter}} \approx \frac{e^2}{4\pi \hbar} \ln \left[\frac{2|\mu_c| - (\omega + i2\Gamma)\hbar}{2|\mu_c| + (\omega + i2\Gamma)\hbar} \right] \quad (3)$$

Where e is the electron charge, K_B is the Boltzmann constant, and $\hbar$ is the reduced Planck constant. The real and imaginary components of the intraband conductivity are plotted as a function of frequency for various chemical potentials μ_c ranging from 0 to 0.4 eV, with a temperature of $T = 300$ K and relaxation time $\tau = 10^{-13}$ s. These results are presented in Figs. 1(a) and (b).

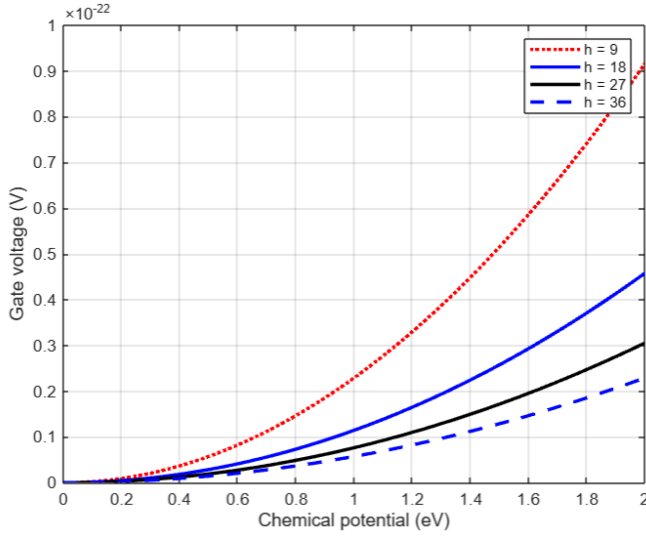


Fig. 2. The variation of gate voltage with respect to chemical potential.

B. Gate Voltage

Chemical doping and electrical gating stand out as the primary methods for adjusting the Fermi level in graphene. By adjusting the voltage (V_g), as indicated in (4), the chemical potential (μ_c) of the graphene layer can be manipulated, consequently impacting the Fermi level [26].

$$V_g = V_0 + \frac{e\mu_c^2 h}{\pi h^2 V_f^2 \epsilon_0 \epsilon_r} \quad (4)$$

In this context, V_0 is the voltage adjustment correlated with chemical doping, the variable h refers to the thickness of the substrate, ϵ_r denotes the relative electric permittivity of the substrate, and ϵ_0 signifies the electric permittivity of free space. Meanwhile, V_f represents the Fermi velocity of graphene, which is specified as $v_f = 9.5 \times 10^5$ m/s. Additionally, Fig. 2 illustrates the gate voltage plotted against the chemical potential (μ_c) for various substrate thicknesses (h).

C. Proposed Biosensor Design

Metamaterials, with their unique properties and ability to manipulate electromagnetic waves, can act as perfect absorbers [36]. These materials are designed with specific structures that enable them to absorb electromagnetic waves, including Terahertz (THz) waves, with high efficiency. When THz waves interact with metamaterials, they can be absorbed, reflected, or transmitted. The absorption of THz waves by metamaterials is crucial for biosensing applications, as it allows for the detection of subtle changes in the properties of biological samples [37].

In the context of biosensing, achieving accurate cancer cell diagnosis requires a biosensor that balances sensitivity, specificity, and selectivity. Overcoming these challenges is pivotal in the design of our proposed work. However, despite these challenges, optimizing biosensor performance is critical to enhance clinical utility and provide valuable insights for cancer detection. The

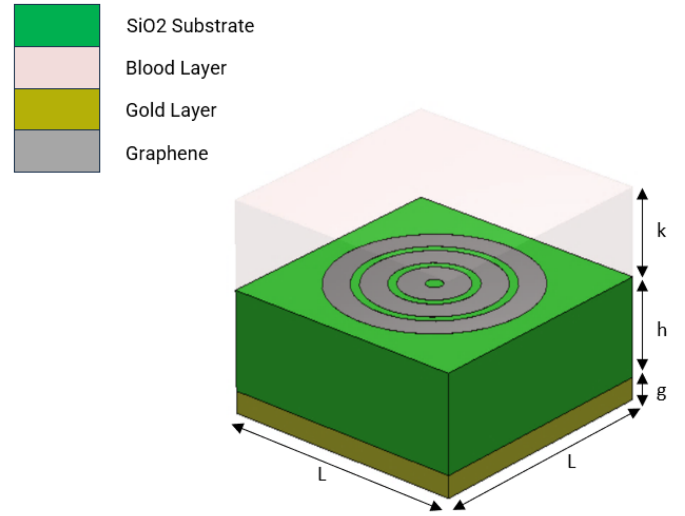


Fig. 3. The perspective view of the proposed graphene-based biosensor.

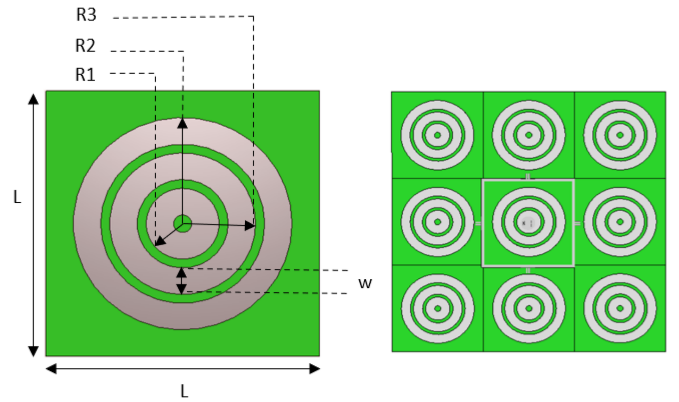


Fig. 4. Top view and periodic array of the suggested structure.

proposed perfect absorber metamaterial, illustrated in Fig. 3, serves as a biosensor in this context. As depicted in the figure, the 3D representation of the proposed metamaterial biosensor's unit cell is composed of three distinct layers: a 2 μm thick Au fully reflective layer, a 9 μm thick SiO2 layer, and a monolayer graphene functioning as an equivalent 2D impedance sheet. The unit cells exhibit a period of $P_x = P_y = 30 \mu\text{m}$. As shown in Fig. 4, the graphene monolayer, with its high surface area and conductivity, consists of three equally spaced rings of varying sizes. The fully reflective layer is characterized by a conductivity of $\sigma = 4.5 \times 10^7$ S/m. The dielectric constant of the intermediate layer is $\epsilon_r = 3.8$, indicating its ability to influence the electromagnetic properties within the structure. Table I showcases the optimized geometric properties of the unit cell.

The objective of this paper is to develop an innovative terahertz biosensor that can discriminate between different types of healthy cells and cancer cells. Therefore, graphene is selected as the most appropriate material. There are four main reasons for selecting graphene as a metasurface for the identification of cells: firstly, graphene has a high conductivity [39], which

TABLE I
THE OPTIMAL DIMENSIONS OF THE PROPOSED BIOSENSOR

Parameter	Value (μm)	Parameter	Value (μm)
L	30	w	3
R1	12	h	9
R2	8	g	2
R3	4	-	-

is essential for efficient biosensor operation. This high conductivity enables graphene to rapidly respond to changes in the electromagnetic field, which is critical for detecting different cell types. Secondly, the tunability of graphene enables it to function as a metasurface adaptable to respond to particular wavelengths of electromagnetic radiation, including terahertz radiation [40]. Thirdly, graphene is a biocompatible material, which means that it is safe for use in biological systems [41]. This biocompatibility is crucial for the development of a biosensor that can be used for the identification of cells in a biological environment. Fourthly, Graphene is considered a cost-effective material for various applications, including biosensing, due to its relatively low production costs and scalability.

The finite-difference time-domain method (FDTD) stands as a potent numerical method for resolving electromagnetic field equations within the time domain [38]. It finds extensive application in diverse fields, including optics, photonics, and electromagnetic compatibility studies, to analyze and optimize the behavior of complex electromagnetic systems. The FDTD method is established on the concept of discretizing space and time into a grid of discrete points and intervals, respectively [42]. Maxwell's equations are discretized and then solved iteratively at each point within the computational grid in the finite-difference time-domain (FDTD) method. This iterative process involves updating the field values at each time step, enabling precise simulation of the propagation and interaction of electromagnetic fields within the defined space [43]. FDTD's capability to capture both spatial and temporal characteristics of fields enhances its utility in electromagnetic simulations. Consequently, it serves as a valuable tool for studying a wide range of applications, including sensor design, wave propagation, and electromagnetic compatibility analysis. In this context, Maxwell's equations are utilized to numerically compute the proposed structure, as denoted by (5) and (6).

$$\frac{\partial \mathbf{E}}{\partial t} = \frac{1}{\epsilon_0} \nabla \times \mathbf{H} \quad (5)$$

ϵ_0 represents the permittivity of free space, and H is the magnetic field intensity.

$$\frac{\partial \mathbf{H}}{\partial t} = -\frac{1}{\mu_0} \nabla \times \mathbf{E} \quad (6)$$

μ_0 represents the permeability of free space, and E is the electric field intensity. The analysis was carried out employing CST Microwave Studio, leading to the calculation of spectral curves within the terahertz frequency spectrum. Before commencing the simulation, the boundary conditions are set as the unit cells in x and y-directions are adopted as periodic, and the z-direction is open boundary for perfect matching. The incident THz wave

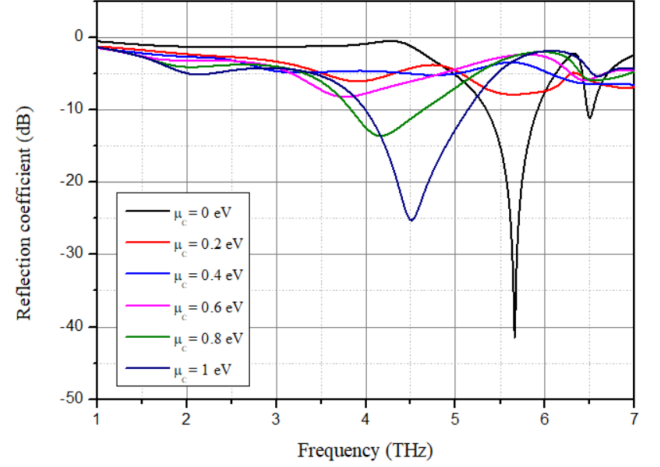


Fig. 5. Influence of chemical potential variation in graphene-based sensor.

in the z-direction for TM and TE waves is modeled as a Floquet port. Additionally, a tetrahedral mesh is used.

III. RESULTS AND DISCUSSION

A. Parametric Analysis

Studying the influence of graphene parameters on surface conductivity is a crucial aspect of designing and analyzing a THz Metamaterial biosensor for detecting cancer cells. By understanding the relationship between the graphene parameters and the surface conductivity, it is possible to optimize the sensor's sensitivity, selectivity, and overall diagnostic accuracy. The graphene parameters, such as the Fermi level, relaxation time, and temperature, can be adjusted using chemical doping or electrical gating. This will be further explored layers in the following investigation by calculating S-parameters.

The graphene-based sensor presented in this work demonstrates highly adaptable performance characteristics, as illustrated in Fig. 5. By adjusting the chemical potential of the embedded graphene layer from 0 to 1 eV, the resonance frequency shifts from $f = 5.6$ to $f = 4.5$ THz. It's important to emphasize that in this study, significant conductivity is attained through the deposition of non-doped graphene, marking a notable outcome. Furthermore, the structure shows limited absorption capabilities in the absence of a graphene layer.

A crucial factor in defining the characteristics and behavior of graphene is its relaxation time, which is one of the most fascinating properties of this material. Fig. 6 illustrates how the resonance frequency of the biosensor changes with variations in the relaxation time under no load conditions. The figure demonstrates that as the relaxation time (τ) of graphene varies from 0.1 ps to 1 ps, the resonance frequency position shifts. Specifically, when the relaxation time is 0.1 ps, the resonance frequency is positioned at 5.66 THz, with the reflection coefficient at -41.45 dB. However, with a relaxation time of 1 ps, the reflection coefficient decreases, reaching -30.7 dB at 5.71 THz. Setting the relaxation time to 0.1 ps seems a critical parameter to enhance the performance of the biosensor. This choice is

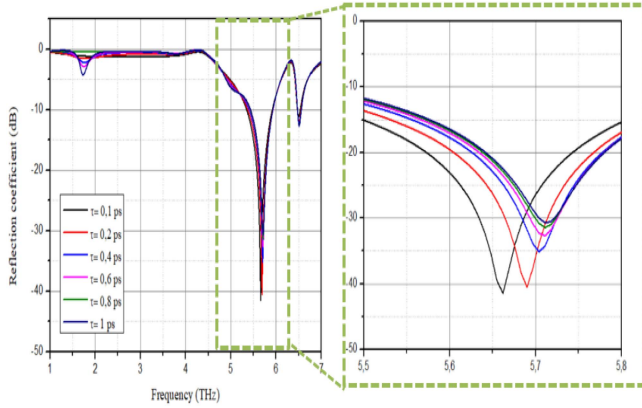


Fig. 6. Impact of relaxation time variation in graphene-based sensor.

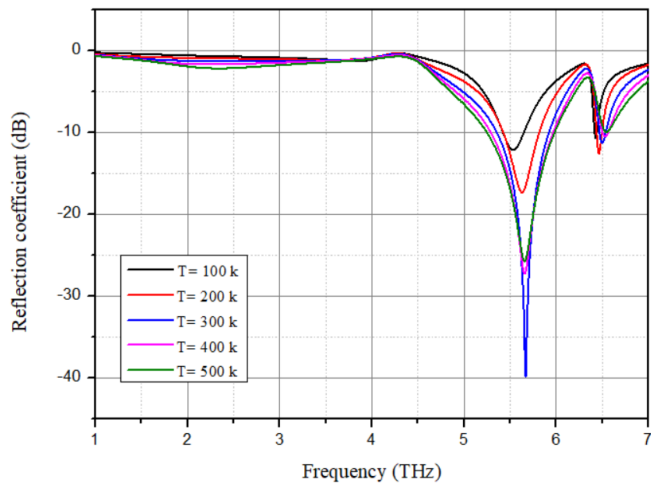


Fig. 7. Influence of temperature variation in graphene-based sensor.

crucial as it has the potential to improve sensitivity and overall functionality.

Additionally, the temperature variation is depicted in Fig. 7. It's apparent that temperature (T) is a variable in (2). As a result, this parameter affects conductivity, consequently influencing the biosensor's performance. From Fig. 7, it's clear that with temperature ranging from 100 K to 500 K, the optimal value appears to be 300 K, resulting in enhanced resonance frequencies and according to best-achieved impedance matching.

The effectiveness of the proposed biosensing platform is evaluated by its ability to detect changes in the refractive index of the surrounding medium. To support the biosensor's suitability for cell investigation, various simulations were conducted using various biological samples. These include pure blood, healthy and cancerous Bse1 skin cells, Jurkat blood cells, HeLa cervical cells, Adrenal gland cells, and MCF-7 Breast cells.

B. Sensor Parameters

To analyze the effectiveness of the proposed biosensing platform, several metrics can be used to provide a comprehensive assessment of its efficiency. These metrics can include sensitivity

(S), figure of merit (FOM), and quality factor (Q) [45]. The quantification of sensitivity involves assessing the ratio between alterations in wavelength and variations in refractive index [44], as indicated by :

$$\text{Sensitivity} = \frac{\Delta f}{\Delta n} (\text{THz/RUI}) \quad (7)$$

Here, Δf represents the frequency change of the resonance peak, and Δn denotes the change in refractive index [46]. The figure of merit (FOM) is defined as the ratio between sensitivity and the full width at half maximum (FWHM) of the resonance peak, which describes the sharpness of the resonance curve.

$$\text{Figure of Merits} = \frac{\text{Sensitivity}}{\text{Full-Width at Half-Maximum}} (\text{RUI}^{-1}) \quad (8)$$

The quality factor (Q) is determined by the ratio of the resonance frequency to the FWHM, serving as a measure of the resonance's quality and its capability to discern alterations in the surrounding medium.

$$\text{Quality - factor} = \frac{f_{\text{res}}}{\text{Full - Width at Half - Maximum}} \quad (9)$$

The full width at half maximum (FWHM) is linked to the quality factor (Q-factor), with a higher Q-factor associated with a lower FWHM value, deriving in an enhanced figure of merit (FOM). consequently, a sensor with a higher Q-factor is desirable for achieving a higher FOM.

C. Sensing Performance

When evaluating the sensitivity of a biosensor for blood concentration detection [47], the focus is on quantitative detection to ensure accurate and reliable results. In this context, conducting precise simulations using pure blood samples at extremely low concentrations becomes essential for testing the biosensor's sensitivity. The blood sample will spread across the upper surface of the biosensor, creating a layer with a thickness of $k = 8 \mu\text{m}$.

Fig. 8 illustrates the sensor's frequency response across various blood densities. The dielectric constant varies with different concentrations of pure blood, resulting in various frequency responses. When the biosensor is loaded with a sample, The lowest and highest frequency shifts correspond to 100% and 25% concentrations, respectively, with the resonance frequencies of these concentrations being 4.955 and 5.662 THz.

In the context of blood concentration detection, the sensitivity of a biosensor may vary depending on the concentration range being tested. Typically, biosensors are designed to exhibit high sensitivity across a wide range of concentrations. For instance, the design of biosensors with specific sensing materials and optimized detection methods can enhance sensitivity and improve the sensor's ability to detect varying blood concentrations effectively. Table II demonstrates that the sensitivity is 0.559 at a concentration of 25%. As the concentration increases, the sensitivity also increases, reaching 0.697 at 100% concentration.

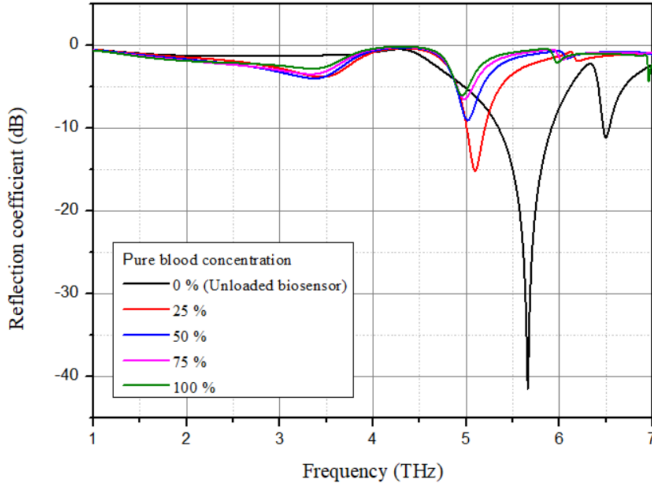


Fig. 8. Simulation results of the graphene-based biosensor loaded with pure blood.

TABLE II
COMPARISON OF THE SENSITIVITY WITH DIFFERENT VALUES OF BLOOD CONCENTRATION

Blood concentration %	Resonance frequency (THz)	Return loss(dB)	Sensitivity (THz/RIU)
Air	5.662	-41.455	-
25 %	5.095	-15.167	0.559
50 %	5.018	-9.074	0.635
75 %	4.983	-6.507	0.669
100 %	4.955	-6.097	0.697

The detection process in this method is based on the scattering parameter and the difference in refractive indices (RIs) between analytes present on the surface of the sensor. Specifically, Terahertz metamaterials' resonance frequency can be likened to the vibrational frequencies inherent in bio-molecules. In the absence of any analyte, the sensor's resonance frequency reflects the environmental refractive index ($n=1$). The primary objective of this work is to discern cellular constituents based on their refractive indices (RIs) and reflection coefficients, as cancerous tissues exhibit distinct properties compared to normal tissues. Specifically, cancerous tissues have higher water content, more capillaries, more types of proteins, and larger nuclei, resulting in a higher RI compared to normal tissues [48]. Therefore, by analyzing the resonance frequency fingerprint, the biosensor can differentiate between normal and cancerous in various organs, even in the early stages of cancer development. In this study, the parameters are set as follows: $T = 300$ K, $\tau = 0.1$ ps and $\mu_c = 0.1$ eV. Fig. 9 depicts how the reflection coefficient varies across different frequencies for both cancer cells and their corresponding normal cells.

In Fig. 9(a), it's evident that the basal cancer type exhibits a resonance frequency of 4.248 THz, while the normal cell resonates at $f_{res} = 4.262$ THz. Consequently, quality factors of 8.935 THz and 8.846 THz, along with sensitivities of 3.880 THz/RIU and 3.721 THz/RIU, are achieved for normal and cancer cells, respectively. Additionally, the resonance frequencies of Jurkat blood cancer and normal cells are found

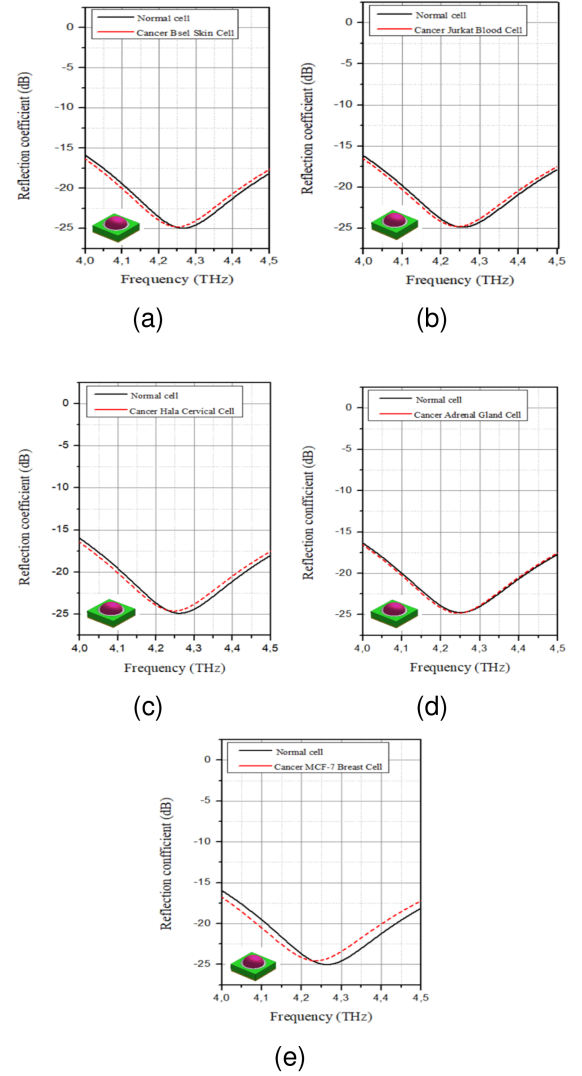


Fig. 9. Variation of the reflection coefficient with the frequency for different types of cancer cells and their corresponding normal cells: (a) Basal skin cell, (b) Jurkat blood cell, (c) HeLa cervical cell, (d) Adrenal gland cell, and (e) MCF-7 Breast cell.

to be 4.241 and 4.255 GHz, respectively, with sensitivities of 3.643 and 3.742 THz/RIU, as shown in Fig. 9(b). Moreover, it is evident from Fig. 9(c) and (d) that sensitivities of 3.607 THz/RIU and 3.597 THz/RIU are achieved for HeLa cervical cell and adrenal gland cell cancer types, respectively, with frequency shifts of 0.014 THz and 0.007 THz calculated, respectively. Furthermore, the analysis of MCF-7 cells yields a quality factor of 8.948 for normal cells and 8.784 for cancer cells, with corresponding sensitivities of 3.880 THz/RIU and 3.560 THz/RIU, as illustrated in Fig. 9(e). The results we obtained are also presented in Table III.

In Fig. 10, the second phase of the investigation focused on elucidating the correlation between the resonance frequency of the sensor and the refractive indices (RI) of both normal and cancerous cells.

We noted distinct declines in transmittance at specific refractive index values for each cell, suggesting resonance or specific

TABLE III
PROPERTIES OF THE BIOSENSOR IN THE PRESENCE OF BOTH HEALTHY AND CANCER CELLS

Cell name	State	Resonance frequency (THz)	Return loss (dB)	Sensitivity (THz/RIU)	FWHM (THz)	FOM (RIU ⁻¹)	Quality factor
Bsel skin cell	Normal	4.262	-24.959	3.880	0.477	8.137	8.935
	Cancer	4.248	-24.800	3.721	0.480	7.748	8.846
Jurkat blood cell	Normal	4.255	-24.810	3.742	0.479	7.808	8.879
	Cancer	4.241	-24.802	3.643	0.480	7.588	8.833
HeLa cervical cell	Normal	4.262	-24.880	3.571	0.478	7.470	8.916
	Cancer	4.248	-24.681	3.607	0.483	7.467	8.795
Adrenal gland cell	Normal	4.248	-24.740	3.711	0.481	7.715	8.831
	Cancer	4.241	-24.820	3.597	0.478	7.525	9.368
MCF-7 Breast cell	Normal	4.262	-25.005	3.880	0.476	8.146	8.948
	Cancer	4.234	-24.550	3.560	0.482	7.385	8.784

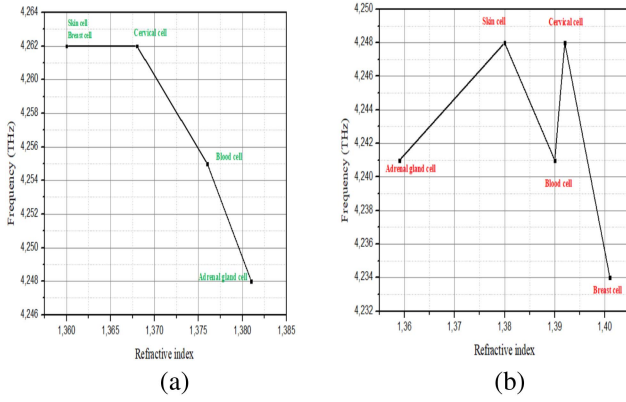


Fig. 10. Resonance frequency of the sensor versus the refractive indices.

absorption of particular light frequencies as they passed through these materials. Upon analyzing the data, we established an approximately linear correlation between the resonance frequency and refractive indices for normal cells, and the corresponding equation is as follows.

$$f_1 - 0.007 = f_2 \quad (10)$$

This equation provides a mathematical representation of how changes in shifts in the resonance frequency with increasing RI. However, the relationship between resonance frequency and refractive indices indicates that with increasing RI, the resonance frequency rises in certain cancer cells and declines in others. This occurrence is attributed to differences in the water content among various cells.

The investigation focused on the electric field within the 5.66 THz region, as depicted in Fig. 11. The field is represented by two horizontal half rings, signaling a potent electric field responsible for absorbing electromagnetic waves within the 5.66 THz range. Fig. 11 demonstrates that the proposed sensor exhibits exceptional performance for biomedical sensor applications, owing to its elevated electric field density. These

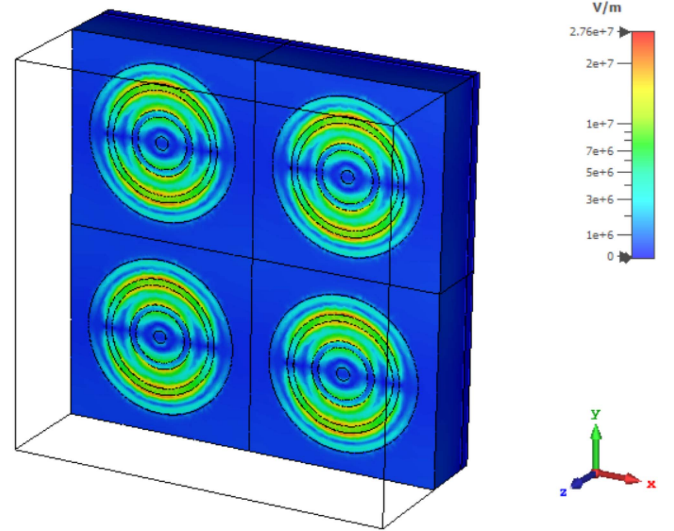


Fig. 11. E-field distributions of the metamaterial structure at 5.66 THz.

field distributions offer insights into the resonance process and aid in comprehending the design's performance.

Examining the intensities of electric and magnetic fields across different frequencies reveals the distinct contributions of specific components to signal resonances across the spectrum. This analysis of characteristics helps fine-tune the design to target specific resonance frequencies more effectively, thereby enhancing the overall performance of the biosensor. Another aspect under investigation involves examining the distribution of the magnetic field. Fig. 12 reveals a highly concentrated magnetic field within a distinct vertical half-ring region, indicating the remarkable sensitivity of the proposed sensor within the 5.66 THz range. This strong magnetic field effectively maintains the polarization angle unchanged, showcasing the exceptional quality of the suggested sensor for biomedical imaging and biosensor applications.

To better understand how the proposed metamaterial biosensor absorbs physical energy, we examined the distribution of

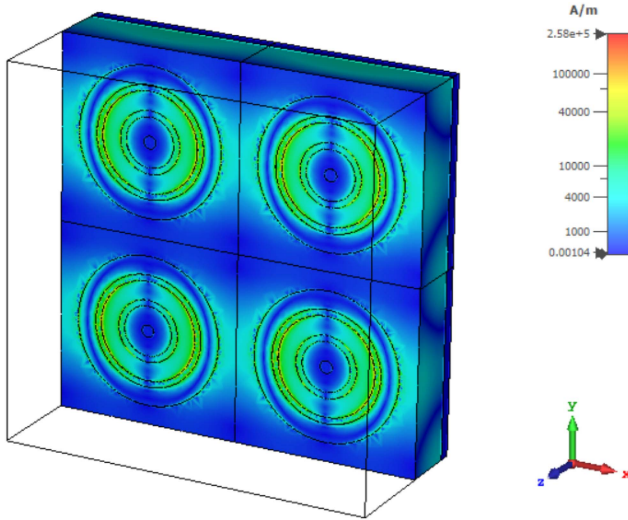


Fig. 12. H-field distributions of the metamaterial structure at 5.66 THz.

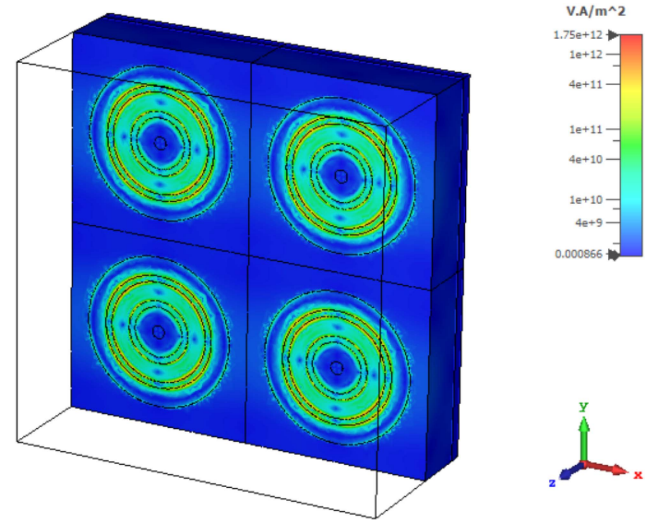


Fig. 14. The power flow distribution of the metamaterial biosensor at 5.66 THz.

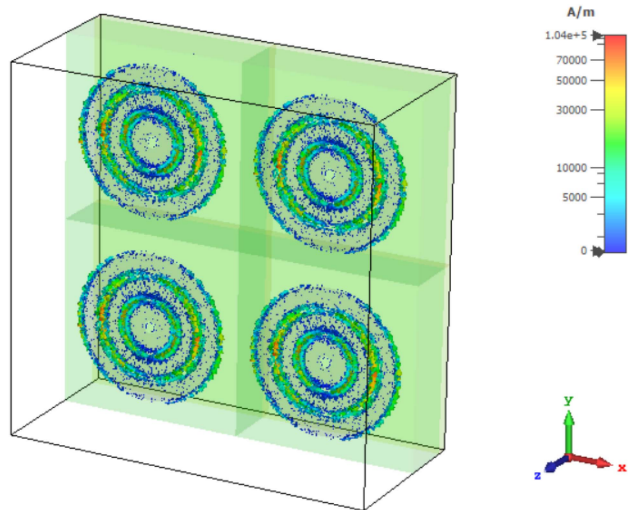


Fig. 13. The surface current distribution of the metamaterial biosensor at 5.66 THz.

surface currents within the top layers of the suggested design. As depicted in Fig. 13, circular parallel currents indicate a strong magnetic response. The distribution of electric and magnetic fields in a given space is directly influenced by the distribution of current flows within that space. When current flows are parallel, the electric and magnetic fields are arranged in a specific pattern. Similarly, when current flows are antiparallel, the fields are arranged differently.

The internal magnetic field created by parallel current flow acts in opposition to the external H-field. As depicted in Fig. 14, it also displays a densely packed region with a resonance frequency of 5.66 THz, indicating a substantial flow of power in a completely circular pattern. The circular power distribution indicates the effectiveness of the proposed sensor design in capturing and utilizing incoming electromagnetic waves.

A biosensor utilizing metamaterials was showcased for detecting penicillin, achieving a quality factor of 5.58 and a

sensitivity of 0.02432 THz/RIU. Additionally, a distinct biosensor was engineered with a focus on detecting breast cancer, showcasing a quality factor of 2.43 and a sensitivity of 1.21 THz/RIU. Moreover, an additional biosensor was designed for virus detection, exhibiting a sensitivity of 0.0242 THz/RIU, although its quality factor was not provided. Furthermore, another study was developed for detecting the analyte, featuring a sensitivity of 0.0203 THz/RIU, a figure of merit ranging between 1.5 and 1.8 RIU⁻¹, and a quality factor of 8.2. Also, a study conducted in 2023 focused on the detection of non-melanoma skin cancer, achieving a sensitivity range of 0.0515 to 0.076 THz/RIU. The biosensor's figure of merit falls between 0.86 and 1.15 RIU⁻¹, with a quality factor of 12.8. To highlight the excellence of the proposed metamaterial biosensor compared to previous similar designs, a comprehensive comparison is presented in Table IV. It's apparent that the current design exhibits higher sensitivity compared to all previous designs and attained 3.880 THz/RIU in the detection of Basel skin cells. In addition, the suggested design has been subjected to testing the detection of MCF-7 Breast cells, and a maximum FOM was achieved with 8.146 RIU⁻¹. Notably, our design exhibits a significantly higher Q-factor across all bands compared to previous biosensors with 8.948, indicating potentially superior performance in distinguishing healthy and cancer cells. This indicates that our proposed biosensor guarantees conclusive results in such differentiation. The quality factor (Q-factor), figure of merit (FOM), and sensitivity are fundamental parameters in determining biosensor performance, with higher values indicating enhanced detection capabilities and accuracy.

The high sensitivity and specificity of THz imaging techniques make them a valuable tool in the field of cancer detection and diagnosis. The ability to differentiate between cancerous and normal tissues based on their THz frequency signatures provides a non-invasive and precise approach to early cancer detection, allowing for timely medical intervention and improved patient outcomes.

TABLE IV
COMPARISON OF BIO-SENSING PERFORMANCE OF DIFFERENT SENSOR APPLICATIONS EMPLOYING THz METAMATERIALS

Ref.	S (THz/RIU)	Q	FOM (RIU ⁻¹)	Year	Bio-application
[49]	0.02432	5.58	0.1216	2014	Detection of Penicillia
[51]	0.0242	-	-	2017	Detection of virus
[50]	1.21	2.43	2.75	2022	Detection of breast cancer
[52]	0.203	8.21-6.05	1.57-1.81	2022	Detection of analyte
[53]	0.0515-0.076	12.8	0.86-1.15	2023	Detection of non-melanoma skin cancer
This work	3.880	8.948	8.146	2024	Detection of Bse1 skin cell, Jurkat blood cell, HeLa cervical cell, Adrenal gland cell, and MCF-7 Breast cell.

IV. CONCLUSION

This article introduces a new graphene-based terahertz biosensor for early detection and diagnosis of healthy and cancerous cells including skin, blood, cervical, adrenal gland, and breast cancer. The absorber has a resonance frequency at 5.662 THz, it has a reasonable frequency shift to different types of cells variations that cover the surface of the absorber. Moreover, implementing the recommended design for cancer detection yields a remarkable sensitivity of 3.880 THz/RIU, coupled with a high-quality factor of 8.948. Furthermore, the proposed design showcases exceptional linearity throughout the entire study range and proves robustness against fabrication variations. Additionally, manufacturing this biosensor can be accomplished using straightforward technologies previously utilized in similar designs.

REFERENCES

- [1] J. Ferlay et al., "Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries," *CA: Cancer J. Clinicians*, vol. 74, no. 3, pp. 229–263, 2024.
- [2] M. Y. Azab, M. F. O. Hameed, A. M. Nasr, and S. S. A. Obayya, "Highly sensitive metamaterial biosensor for cancer early detection," *IEEE Sensors J.*, vol. 21, no. 6, pp. 7748–7755, Mar. 2021.
- [3] Gul-E-Saba Chaudhry et al., "Cancer and disease diagnosis - biosensor as potential diagnostic tool for biomarker detection," *J. Adv. Pharmaceut. Technol. Res.*, vol. 13, pp. 243–247, 2022.
- [4] S. Kiani et al., "Band-stop filter sensor based on SIW cavity for the non-invasive measuring of blood glucose," *IET Wireless Sensor Syst.*, vol. 9, pp. 1–5, 2019.
- [5] X. Sun et al., "Material-enhanced biosensors for cancer biomarkers detection," *Microchemical J.*, vol. 194, 2023, Art. no. 109298.
- [6] M. H. Son et al., "Recent advances in electrochemical and optical biosensors for cancer biomarker detection," *BioChip J.*, vol. 17, pp. 44–67, 2023.
- [7] A. Pawar et al., "Terahertz technology and its applications," *Drug Invention Today*, vol. 5, pp. 157–163, 2013.
- [8] H.-B. Liu, H. Zhong, N. Karpowicz, Y. Chen, and X.-C. Zhang, "Terahertz spectroscopy and imaging for defense and security applications," *Proc. IEEE*, vol. 95, no. 8, pp. 1514–1527, Aug. 2007.
- [9] B. Ferguson and X.-C. Zhang, "Materials for terahertz science and technology," *Nature Mater.*, vol. 1, pp. 26–33, 2002.
- [10] C. Liu, C. Wang, and J. C. Cao, "Multipath propagation channel modeling and capacity analysis for terahertz indoor communications," *J. Opt. Technol.*, vol. 84, pp. 53–61, 2017.
- [11] H. Cheon, H.-J. Yang, and J.-H. Son, "Toward clinical cancer imaging using terahertz spectroscopy," *IEEE J. Sel. Topics Quantum Electron.*, vol. 23, no. 4, Jul/Aug. 2017, Art. no. 8600109.
- [12] J. Chen, S. Hu, S. Zhu, and T. Li, "Metamaterials: From fundamental physics to intelligent design," *Interdiscipl. Mater.*, vol. 2, pp. 5–29, 2022.
- [13] O. Alsalman et al., "Design of split ring resonator graphene metasurface sensor for efficient detection of brain tumor," *Plasmonics*, vol. 19, pp. 523–532, 2024.
- [14] S. Asgari and T. Fabritius, "Multi-band terahertz anisotropic metamaterial absorber composed of graphene-based split square ring resonator array featuring two gaps and a connecting bar," *Sci. Rep.*, vol. 14, pp. 74–77, 2024.
- [15] V. B. Mbayachi et al., "Graphene synthesis, characterization and its applications: A review," *Results Chem.*, vol. 3, 2021, Art. no. 100163.
- [16] A. A. Balandin, "The heat is on: Graphene applications," *IEEE Nanotechnol. Mag.*, vol. 5, no. 4, pp. 15–19, Dec. 2011.
- [17] M. Jablan, M. Soljčić, and H. Buljan, "Plasmons in graphene: Fundamental properties and potential applications," *Proc. IEEE*, vol. 101, no. 7, pp. 1689–1704, Jul. 2013.
- [18] Z. Liu, H. Zhang, S. Liu, B. Bian, and X. Kong, "An ultra-thin polarization-insensitive wide-angle metamaterial absorber," in *Proc. IEEE Prog. Electromagnetics Res. Symp. - Spring*, 2017, pp. 1887–1892.
- [19] Q. A. Alsulami et al., "A tunable and wearable dual-band metamaterial absorber based on polyethylene terephthalate (PET) substrate for sensing applications," *Polymers*, vol. 14, Oct. 2022, Art. no. 4503.
- [20] H. Wakatsuchi, D. F. Sievenpiper, and C. Christopoulos, "Designing flexible and versatile metamaterial absorbers," *IEEE Electromagn. Compat. Mag.*, vol. 5, no. 2, pp. 76–82, Secondquarter 2016.
- [21] X. Lu, H. Ge, Y. Jiang, and Y. Zhang, "Design and analysis of a dual-band THz metamaterial sensor with high refractive index sensitivity," *Front. Phys.*, vol. 10, pp. 1–10, 2022.
- [22] S. J. Mishu, M. A. Rahman, and N. Dhar, "Highly sensitive refractive index sensing with a dual-band optically transparent ITO-based perfect metamaterial absorber for biomedical applications," *Heliyon*, vol. 10, no. 5, Feb. 2024, Art. no. e26842.
- [23] M. Xu, T. Liang, M. Shi, and H. Chen, "Graphene-like two-dimensional materials," *Chem. Rev.*, vol. 113, no. 5, pp. 3766–3798, 2013.
- [24] C. Nie et al., "Recent progresses in graphene based bio-functional nanostructures for advanced biological and cellular interfaces," *Nano Today*, vol. 29, pp. 57–97, 2019.
- [25] M. Rezeg, A. Hlali, A. Oueslati, and H. Zairi, "A novel approach utilizing graphene-based microfluidic technology for skin cancer detection," *Prog. Electromagnetics Res. C*, vol. 141, pp. 89–100, 2024.
- [26] A. Hlali, A. Oueslati, and H. Zairi, "Numerical simulation of tunable terahertz graphene-based sensor for breast tumor detection," *IEEE Sensors J.*, vol. 21, no. 8, pp. 9844–9851, Apr. 2021.
- [27] F. Kazemi, "High Q-factor compact and reconfigurable THz aperture antenna based on graphene loads for detecting breast cancer cells," *Superlattices Microstructures*, vol. 153, 2021, Art. no. 106865.
- [28] Z. Vafapour, W. Troy, and A. Rashidi, "Colon cancer detection by designing and analytical evaluation of a water-based THz metamaterial perfect absorber," *IEEE Sensors J.*, vol. 21, no. 17, pp. 19307–19313, Sep. 2021.
- [29] H. Sharifi and S. Eskandari, "Sensing blood components and cancer cells with photonic crystal resonator biosensor," *Results Opt.*, vol. 14, 2024, Art. no. 100593.
- [30] D. Li et al., "Identification of early-stage cervical cancer tissue using metamaterial terahertz biosensor with two resonant absorption frequencies," *IEEE J. Sel. Topics Quantum Electron.*, vol. 27, no. 4, Jul/Aug. 2021, Art. no. 8600107.
- [31] R. H. Jibon, E. Podder, and A. A.-M. Bulbul, "Adrenal glands cancer detection using PCF-based biosensor," in *Proc. IEEE Int. Conf. Electron., Commun. Inf. Technol.*, 2021, pp. 1–4.
- [32] D. S. Sitnikov et al., "Effects of high intensity non-ionizing terahertz radiation on human skin fibroblasts," *Biomed. Opt. Exp.*, vol. 12, no. 11, pp. 7122–7138, Oct. 2021.

- [33] Y. Zhang et al., "Continuous-wave THz imaging for biomedical samples," *Appl. Sci.*, vol. 11, no. 1, 2020, Art. no. 71.
- [34] T. Mohiuddin, E. Hill, D. Elias, A. Zhukov, K. Novoselov, and A. Geim, "Graphene in multilayered CPP spin valves," *IEEE Trans. Magn.*, vol. 44, no. 11, pp. 2624–2627, Nov. 2008.
- [35] G. D. Bouzianas, N. V. Kantartzis, T. V. Yioultsis, and T. D. Tsiboukis, "Consistent study of graphene structures through the direct incorporation of surface conductivity," *IEEE Trans. Magn.*, vol. 50, no. 2, Feb. 2014, Art. no. 7003804.
- [36] R. Zhou et al., "3D printing metamaterials for highly efficient electromagnetic wave absorption," *Sci. China Mater.*, vol. 66, pp. 1283–1312, 2023.
- [37] W. Xu et al., "Terahertz biosensing with a graphene-metamaterial heterostructure platform," *Carbon*, vol. 141, pp. 247–252, 2018.
- [38] F. L. Teixeira et al., "Finite-difference time-domain methods," *Nature Rev. Methods Primers*, vol. 3, 2023, Art. no. 75.
- [39] H. Murata et al., "High-electrical-conductivity multilayer graphene formed by layer exchange with controlled thickness and interlayer," *Sci. Rep.*, vol. 9, 2019, Art. no. 4068.
- [40] Y. Zhang et al., "Direct observation of a widely tunable bandgap in bilayer graphene," *Nature*, vol. 459, pp. 820–823, 2009.
- [41] S. Syama and P. V. Mohanan, "Safety and biocompatibility of graphene: A new generation nanomaterial for biomedical application," *Int. J. Biol. Macromolecules*, vol. 86, pp. 546–55, 2016.
- [42] P. Pietruszka, T. P. Stefański, and J. Gulgowski, "FDTD method for electromagnetic simulations in media described by time-fractional constitutive relations," *Appl. Sci.*, vol. 13, no. 19, 2023, Art. no. 10654.
- [43] D. E. McCoy et al., "Finite-difference time-domain (FDTD) optical simulations: A primer for the life sciences and bio-inspired engineering," *Micron*, vol. 151, 2021, Art. no. 103160.
- [44] M. Gómez-Castño, J. L. García-Pomar, L. A. Pérez, S. Shanmugathan, S. Ravaine, and A. Mihi, "Electrodeposited negative index metamaterials with visible and near infrared response," *Adv. Opt. Mater.*, vol. 8, no. 19, 2020, Art. no. 2000865.
- [45] G. Liu, J. Chen, P. Pan, and Z. Liu, "Hybrid metal-semiconductor metasurface based photoelectronic perfect absorber," *IEEE J. Sel. Topics Quantum Electron.*, vol. 25, no. 3, May/Jun. 2019, Art. no. 4600507.
- [46] C. Cen et al., "A tunable plasmonic refractive index sensor with nanoring-strip graphene arrays," *Sensors*, vol. 18, 2018, Art. no. 4489.
- [47] C. B. Reid, G. Reese, A. P. Gibson, and V. P. Wallace, "Terahertz time-domain spectroscopy of human blood," *IEEE J. Biomed. Health Inform.*, vol. 17, no. 4, pp. 774–778, Jul. 2013.
- [48] E. Di Gregorio, S. Israel, M. Staelens, G. Tankel, K. Shankar, and J. A. Tuszyński, "The distinguishing electrical properties of cancer cells," *Phys. Life Rev.*, vol. 43, pp. 139–188, 2022.
- [49] S. J. Park et al., "Detection of microorganisms using terahertz metamaterials," *Sci. Rep.*, vol. 4, no. 1, May 2014, Art. no. 4988.
- [50] C. Tan et al., "Cancer diagnosis using terahertz-graphene-metasurface-based biosensor with dual-resonance response," *Nanomaterials*, vol. 12, no. 21, Nov. 2022, Art. no. 3889.
- [51] S. J. Park, S. H. Cha, G. A. Shin, and Y. H. Ahn, "Sensing viruses using terahertz nano-gap metamaterials," *Biomed. Opt. Exp.*, vol. 8, no. 8, 2017, Art. no. 3551.
- [52] H. Hu, B. Qi, Y. Zhao, X. Zhang, Y. Wang, and X. Huang, "A graphene based THz metasurface sensor with air-spaced structure," *Front. Phys.*, vol. 10, Sep. 2022, Art. no. 990126.
- [53] M. N. Hamza and M. T. Islam, "Designing an extremely tiny dual-band biosensor based on MTMs in the terahertz region as a perfect absorber for non-melanoma skin cancer diagnostics," *IEEE Access*, vol. 11, pp. 136770–136781, 2023.
- [54] B. Cai, X. Zhu, L. Wu, L. Yang, and Y. Cheng, "Visible quintuple narrowband metasurface absorber based on gallium arsenide square cavity for multispectral sensing application," *IEEE Sensors J.*, vol. 24, no. 12, pp. 19229–19236, 2024.
- [55] Z. Li et al., "Dual-band tunable terahertz perfect absorber based on all-dielectric InSb resonator structure for sensing application," *J. Alloys Compounds*, vol. 925, 2022, Art. no. 166617.
- [56] Z. Huang et al., "High-resolution metalens imaging polarimetry," *Nano Lett.*, vol. 23, pp. 10991–10997, 2023.
- [57] B. Cai et al., "Ultra-broadband and wide-angle plasmonic light absorber based on all-dielectric gallium arsenide (GaAs) metasurface in visible and near-infrared region," *Results Phys.*, vol. 85, 2024, Art. no. 107509.
- [58] C. Rong, B. Cai, Y. Cheng, F. Chen, H. Luo, and X. Li, "Dual-band terahertz chiral metasurface absorber with enhanced circular dichroism based on temperature-tunable InSb for sensing applications," *Phys. Chem. Chem. Phys.*, vol. 26, no. 6, pp. 5579–5588, 2024.
- [59] Z. Ju et al., "Ultra-broadband high-efficiency airy optical beams generated with all-silicon metasurfaces," *Adv. Opt. Mater.*, vol. 9, no. 1, pp. 1–8, 2020.
- [60] M. Deng et al., "Broadband angular spectrum differentiation using dielectric metasurfaces," *Nature Commun.*, vol. 15, 2024, Art. no. 2237.
- [61] L. Wu et al., "Ultra-broadband and wide-angle plasmonic absorber based on all-dielectric gallium arsenide pyramid nanostructure for full solar radiation spectrum range," *Int. J. Thermal Sci.*, vol. 201, 2024, Art. no. 109043.
- [62] D. Wang et al., "Transmission/reflection mode switchable ultra-broadband terahertz vanadium dioxide (VO₂) metasurface filter for electromagnetic shielding application," *Surfaces Interfaces*, vol. 49, 2024, Art. no. 104403.